Restoring Operational Histories with Generated Temporal Graph Supervision

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Abstract

An analyst investigating a purchase order or loan application needs its complete history, including cancellations and retries. Missing object links, wrong joins and fragmented identifiers can hide relevant records or introduce false evidence. We study offline repair of these associations while preserving recorded event content. Executable workflows provide training references; separate observation and fault models vary the conditions under which a learner must recover them. We evaluate the resulting supervision against models trained on the same limited number of verified operational histories, counting both fitting and calibration labels. A separate comparison adapts the author-provided HERE Transformer, without expert rules, to its supported missing-case-identifier task. Evaluation measures retrieved histories, operational queries, false identity merges and newly damaged records. Public business-log experiments use controlled injected faults. Across fresh purchasing histories, a generated-plus-real model with 64 verified bundles achieves mean history F1 of .856, exceeding .845 from real-only fitting with 256 bundles in all three paired trials. Gains depend on the application and operating point: loan improvements are smaller, the purchasing mixture beats generated-only repair in just one trial, and stricter thresholds trade recovery for fewer newly damaged records. A separate missing-ID comparison exposes severe generated-only purchasing failures. Generated supervision can reduce incremental verification requirements in the tested setting, but does not establish reliable automatic repair across processes or fault types.

1 Application and contribution

A cancellation can be correctly recorded yet absent from the history retrieved for an order. A wrong join can make another customer's record appear in that history. When two systems assign different identifiers to the same object, either identifier may retrieve only part of its evidence. These failures make operational investigation difficult even when activities and timestamps themselves are accurate.

The practical objective is to restore recorded evidence, including exceptional behavior. We retain cancellations, returns and repeated work rather than changing the record to resemble an ideal process. The input is an available window of event–object relations, and the output is an auditable set of association and identity edits. This is offline reconstruction; later observations inside the window can support an earlier association.

Training such a repair system requires examples of correct histories and the ways their observed connections can fail. Verification is costly because an investigator may need to reconcile several records and identifiers. Executable workflows offer exact symbolic references, but simulated observations can contain misleading regularities. For example, an amount may be constant throughout a simulated order while the corresponding operational measurement changes between activities. A learner can mistake such differences for incorrect joins.

We therefore ask two linked application questions. Does varying observations of an unchanged workflow execution improve generated repair supervision? When some histories can be verified, does this supervision add value beyond a model using the same verification budget? The second question includes the labels needed to choose operating thresholds. Improved average retrieval is insufficient if a method introduces damaging false associations.

The contribution is an executable training-data protocol and its evaluation for coupled temporal association and identity repair. The classifiers, observation-randomization principle and graph operators are established components. The evidence includes a paired observation study, explicitly accounted adaptation budgets, a published-code comparison on a compatible task, and measurements of useful recovery and harm. The generator is a bounded component; building a universal scientific dataset platform is outside this paper.

2 Related work and comparison scope

HERE combines a Transformer with optional expert knowledge for missing case identifiers [1]. We retain its author model, tensor dataset and greedy decoder in a separate single-case comparison. Its output contract does not directly express simultaneous multi-object participation edits and alias merging. SANAGRAPH repairs event-log attributes using supplied trace structure [2]; providing correct trace membership would expose a target of our task. DDTR and process-tree approaches recover trace content [3, 4]. Their objectives differ from preserving event content while restoring its associations.

Contextual Event Knowledge Graph inference uses available relations and physical rules to infer missing entity identifiers [8]. Object-centric imperfection patterns identify association and identity problems encountered in practice [9]. Their public artifacts include injected event–object faults and observed cross-log object–object disagreements. The latter are useful evidence of data inconsistency, but do not independently certify the missing or wrong event–object links evaluated here. We do not claim to introduce these error categories.

Object-centric simulation [7], configurable process-data faults [10], and temporal graph generation [14] supply relevant building blocks. Domain randomization is an established transfer strategy [11]; synthetic relational pretraining also has recent precedents [12, 13]. Our question concerns what these ideas buy an operational repair application under explicit information and verification limits. HoloClean illustrates joint evidence-based data repair [5]; the present study concentrates on temporal histories, their identity fragmentation and the consequences of repairing them.

3 Task and executable supervision

3.1 Observed information and permitted edits

An event is $e = (u, t, a, P, x)$: an opaque handle, time, activity, typed participants and legitimate descriptive fields. Event–object incidence and temporal adjacency within observed histories form the temporal graph. Equal-time records remain simultaneous; source row order supplies no causal ordering. A typed catalog contains the available object handles, including any aliases.

The system can add or remove participation and merge same-type aliases. It cannot invent events, change recorded activities or times, create entirely unknown entities, or split an already overmerged identity. A merge consolidates handles and deduplicates participation. These capabilities matter when choosing a dataset: natural errors requiring a different operation cannot validate this repair vocabulary.

Only the observed graph and target-type schema enter inference. The canonical identity map, clean case containers, corruption locations, seeds and clean-derived adjacency remain outside the learner interface. Target links are corrupted while resources and, where available, vendors remain context. This is a scoped fault model, not evidence of robustness to unreliable timestamps or pervasive context corruption. Public logs provide reference representations for controlled fault injection; they are not certifications that every underlying business record is correct.

3.2 Generate execution, observation and recording faults separately

Executable worlds contain parents, work items, resources and optional batches. Mechanisms include branching, joins, resource queues, handoffs, multistage routes, repeated work, cancellations and returns. Queueing configurations enforce resource capacity. The worlds are symbolic workflows, not calibrated digital twins of the evaluated organizations.

Observation projection determines visible entity types and fields. Recording faults are applied afterwards, producing learner observations and separate participation/identity references. Normalized recipes, random states, source fingerprints and export hashes record provenance. Execution checkpoints reproduce uninterrupted runs. Observations sharing an execution stay in the same data split.

The paired observation comparison holds executed events, timestamps, target types and faults fixed. Generated views retain original fields or introduce activity-dependent categories, noisy numerical measurements, incidental categories and time bands, with varying field availability. Measurements can depend on an observed amount and activity, but never on a target identifier. Clean and damaged versions receive aligned descriptions through builder-only execution ancestry. Each trial contains 512 fitting worlds and 96 generated calibration worlds. Three independently generated collections test sensitivity to generated training data.

The mixed fault family splits eligible histories at a temporal cut with probability .25, removes approximately .10 of each resulting handle's links in a contiguous block, and replaces surviving target links with probability .10 by a different same-type identity. Histories require at least four events to be fragmented. Missing-block widths use stochastic rounding so short histories can also lose links. A separately implemented pipeline stressor uses activity-routed aliases, a recording outage and persistent stale joins. These are declared experimental conditions, not estimates of natural incident rates.

3.3 Learn and apply auditable edits

Candidate retrieval uses temporal proximity, activity, co-participation and descriptive agreement. Participation features summarize history degree, temporal bracketing, transitions and contextual agreement. The query event is excluded from the evidence supporting its existing association. Identity features compare temporal boundaries and activity, context and field profiles. Opaque identifier values are not numerical predictors.

Separate boosted classifiers score additions, removals and merges. An addition is labelled correct when it supplies a missing reference entity; a removal when the current entity is incorrect; a merge when both handles refer to the same entity. Generated training includes partial teacher-repaired states. These local targets do not guarantee improvement in final history retrieval. Earlier neural controls use the same candidates with MLP, GraphSAGE [6] or temporal message-passing scorers; architecture superiority is not our contribution.

The decoder selects a commuting batch, applies it, recomputes observed evidence and repeats for at most four passes in the revision experiments. It cannot simultaneously merge an entity and

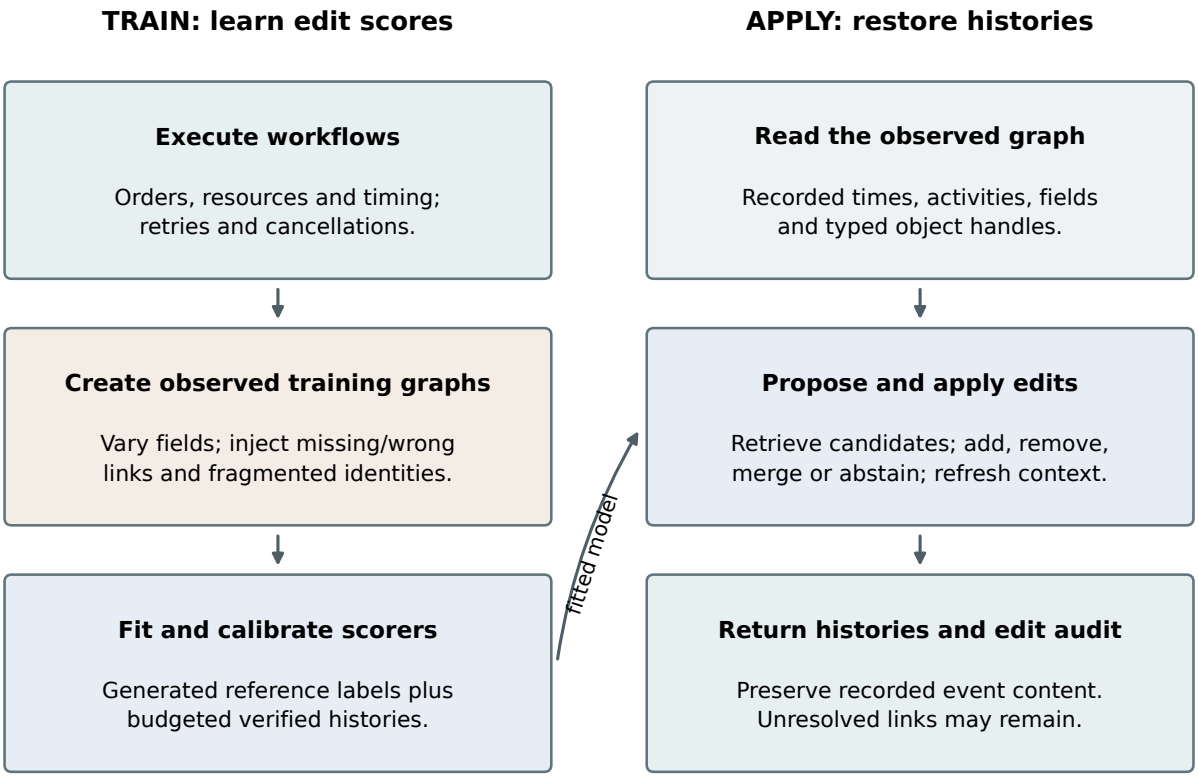


Figure 1: Executable references supervise repair training. Inference uses the observed temporal graph and preserves recorded event content. Reference identities and fault locations remain outside learner input.

edit its participation. Accepted participation edits are not reversed; accepted merges cannot be split by inference. Every batch retains a reversible audit record. For operational graphs, candidate limits are 128 participants per event/type and 64 identity candidates per anchor. Candidate coverage is reported separately from scoring accuracy.

4 Verified-history budgets and evaluation

4.1 Public sources and prospectively assigned cohorts

Loan data come from BPI 2017 [16]. Targets are applications and explicitly referenced offers; resources are context. Purchasing data use the released BPI 2019 OCEL representation [17, 18, 19]; targets are purchase orders and items, with resources and vendors as context. Identifiers and copied trace/object fields are removed from learner attributes. The loan adapter retains allowed event-level descriptions; the purchasing adapter retains cumulative net worth. An absent native offer link is not treated as proof that no real-world relationship existed.

The revision excludes the earlier feasibility prefixes, earlier confirmation roots and their target children. Roots sharing target children, or crossing an excluded target child, are also excluded. A declared hash ranking assigns 512 development, 1,024 annotation and 1,024 confirmation roots per

Source	Bundles	Verified events	Target entities	Alias handles
Loans	16	586–672	36–44	41–54
Loans	64	2,317–2,388	152–159	178–185
Loans	256	9,132–9,533	600–614	703–740
Purchasing	16	204–880	51–178	63–218
Purchasing	64	1,333–1,844	275–398	337–478
Purchasing	256	5,448–5,947	1,170–1,434	1,438–1,725

Table 1: Total simulated verification access, including fitting and calibration. Ranges cover the three declared allocation/fault seeds. A complete bundle includes associated target children and all their aliases. Counts are not measured annotator time; query subsampling does not reduce the reported verification cost.

source, followed by a separate native-method block. Associated target children stay with their root; context resources and vendors may overlap. The native block was prospectively extended from 256 to 2,048 roots for a credible training volume, before its outcomes were inspected. All preceding root partitions were checked to remain unchanged.

The main confirmation cohorts contain 39,421 loan events and 21,680 purchasing events. The native-method cohorts contain 77,981 and 40,483 events. These are fresh object histories from previously studied processes, not new institutions or untouched source domains. Earlier study outcomes inform revision development and are identified separately from the new confirmation.

4.2 Count fitting and calibration together

A verification unit is a complete root history bundle: the root, its target children, all their alias handles and their event memberships. A restricted label view exposes exactly the selected bundles. It cannot reveal the canonical identities of unselected histories. Complete alias verification makes an unlisted candidate a valid negative for a selected entity; this assumption must hold for the simulated verification protocol to represent real annotation.

Total budgets of 16, 64 and 256 bundles allocate 12/48/192 to fitting and 4/16/64 to calibration. Hash-based assignments are nested, and a root retains its role as the budget increases. Three allocation/fault seeds are retained. Table 1 counts verified events, entities and aliases; the artifact also records sampled queries. Each role uses at most 4,096 event queries; the complete verification cost is still counted. These are simulated reference accesses, not measured human work hours.

Real-only, original-generated-plus-real and varied-generated-plus-real classifiers receive identical real labels at each budget. Hyperparameters are fixed: 200 boosting iterations, 15 leaves, learning rate .08, regularization 1 and no early stopping. Mixed fitting initially gives generated and real examples equal weight mass, followed by class balancing. The generated collection is held fixed across these adaptation seeds. If fitting lacks a positive or negative class for an operation, that operation abstains.

Calibration chooses the highest-recall threshold attaining empirical precision of at least .95 and at least five correct actions for each operation. An infeasible operation abstains. A zero-budget control uses generated calibration. A calibration-only control spends the complete selected target budget on thresholds for fixed generated weights, without target fitting. It uses the same selected bundles as the fitting-plus-calibration comparisons. Thus calibration is neither free nor silently supplied from another labelled target set. Historical research and development labels are additional costs; these curves measure incremental adaptation to new histories in known processes.

4.3 Measure useful recovery and introduced errors

For reference entity q, let H_q be its event history, A_q its observed aliases, and $\widehat{H}(u)$ the events retrieved by the predicted cluster of handle u. The primary outcome is

$$F_{\text{history}} = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{|A_q|} \sum_{u \in A_q} \frac{2|\widehat{H}(u) \cap H_q|}{|\widehat{H}(u)| + |H_q|}.$$

Entities with no reference events do not enter Q. Exact-history recovery replaces the inner score by an equality indicator, retaining the same entity/alias weighting. It does not require every alias of an entity to succeed simultaneously.

We separately measure identity-pair precision/recall and changes to initially correct events. The revised damage measure expands each predicted cluster to *all* its reference identities before checking participation. Consequently, a false merge can damage events on both sides, including events whose representative handle did not change. Direct assignment damage is also retained for comparison with the earlier study. Event activity, timestamp and fields are checked to remain unchanged.

Application queries ask whether a retrieved history contains recorded cancellation evidence and whether its first/last timestamps are recovered. Cancellation evidence uses the case-insensitive substring `cancel` consistently in predictions and references. It does not establish the final business outcome, and boundary timestamps do not certify every simultaneous boundary record. Three-seed means and observed ranges summarize the declared trials; events are not treated as independent statistical replications.

4.4 Reliability and published-method controls

The reliability study tests a simple schema control where the public representation declares exactly one root per event. Removing a sole root requires a supported replacement, committed in the same batch. A missing root can receive one supported candidate. Other target types and alias decisions retain the original decoder rules. The middle 64-bundle budget and real-only/varied-mixture regimes are fixed for development comparison. A stricter .99 empirical-precision operating point for the original decoder is essential: a more complex control is useful only if it improves recovery at comparable damage. This control imposes an explicit application schema and does not guarantee correct identities.

The native comparison independently masks each event's single root identifier with probability .30 in each 2,048-root cohort, using three fixed seeds. Activity, resource and time are available; child identifiers, vendor relations and descriptive fields are removed for every method. Trusted remaining case IDs provide closed-case supervision. HERE's author Transformer uses 512-dimensional states, three layers, eight heads, 15-event windows, batch size eight and 30 epochs. The final epoch is evaluated. Only retained identifiers enter training targets; overlapping-window validation is diagnostic, not an independent test. The headless runner and masking/evaluation protocol are explicit adaptations. No expert rules are supplied, so the comparison is named *HERE-Transformer without expert rules*, not a reproduction of the complete published system.

Matched controls include both fixed generated scorers, a target-local boosted scorer and nearest-known-event matching. Generated scorers choose one candidate for each missing ID. The local boosted control fits at most 4,096 retained-ID queries with leave-query-out profiles, whereas HERE uses the remaining known events throughout its training windows; this unequal fitting volume is reported. All retained memberships are observed context. Existing IDs remain unchanged. Native results therefore assess missing-ID completion, not robustness to wrong existing IDs, coupled alias repair or verification-budget efficiency. Data, fixed generated weights, implementation and all

Table 2: History F1 in the earlier frozen paired observation study; mean [min, max] over three confirmation trials.

Domain	Damaged	Original generated	Varied generated
Loan (BPI 2017)	0.7872 [0.7865, 0.7878]	0.7914 [0.7878, 0.7963]	0.8050 [0.8012, 0.8076]
Purchasing (BPI 2019)	0.7598 [0.7561, 0.7640]	0.8091 [0.7915, 0.8247]	0.8263 [0.8114, 0.8412]
BPI 2012 check	0.7453 [0.7413, 0.7524]	0.7760 [0.7662, 0.7825]	0.8155 [0.8069, 0.8274]

six native conditions were pinned before native outcomes; newly fitted checkpoints were recorded afterwards. The native comparison finished before the separate 156-job combined-repair protocol was frozen. That declaration retained both generated mixtures and all budget comparisons, with no subsequent model or threshold selection from confirmation outcomes.

5 Results and application interpretation

5.1 Earlier paired observation study

These results are from the earlier frozen paired observation study, not the fresh revised native confirmation. On BPI 2017, the separate pipeline-fault condition decreased history F1 in all three trials (mean 0.7785 to 0.7771); no true identity pairs were recovered. On the BPI 2012 check partition, source-supervised training reached 0.8778 [0.8692, 0.8887], above the varied-generated mean in the table. A separate generated simulator confirmation over 96 held-out worlds reported original-observation F1 0.7588 [0.7554, 0.7650] and varied-observation F1 0.8000 [0.7913, 0.8113], from a shared damaged baseline of 0.7340.

5.2 Fresh confirmation of combined repair

All 156 declared jobs completed. Table 3 and Fig. 2 compare identical verification budgets on the reserved histories. The original-generated mixture improves loan history F1 by 2.59 percentage points at budget 16 and 1.13 points at budget 64, averaged over the three trials. At budget 256 its mean falls slightly below real-only fitting. The varied mixture loses to real-only fitting in every loan trial at budget 256. Observation variation that helped generated-only transfer therefore does not automatically improve mixed fitting.

Purchasing provides the clearest low-verification result. At budget 64, the varied mixture reaches 85.6% history F1 versus 80.6% for real-only fitting, winning all three paired trials. It also exceeds the 84.5% real-only mean at budget 256 in every paired trial. This comparison uses one quarter as many complete verified bundles, including calibration, under the declared protocol. It does not measure annotation time or remove the historical development cost.

The generated-only controls qualify this finding. With no new target verification, original/varied supervision reaches 79.4/80.8% on loans and 81.9/84.0% on purchasing. The purchasing varied mixture at budget 64 improves on its generated-only counterpart in just one of three trials, despite its higher mean. Moreover, simply recalibrating fixed generated weights is not a substitute for mixed fitting: calibration-only mean F1 is 75.6–81.0% on loans and 80.8–81.7% on purchasing across the three budgets. One 16-bundle loan calibration run damages 9,028 previously correct events. Small calibration sets can produce a permissive operating point that fails when applied repeatedly to a new graph.

Table 3: Fresh combined-repair confirmation. Mean history F1 (percent) over three allocation/fault seeds. Budgets count fitting and calibration together; Fig. 2 shows observed ranges.

Application	Budget	Real only	+ Original	+ Varied	Calibration
Loans	16	82.5	85.1	83.9	75.6
Loans	64	84.2	85.3	83.6	79.2
Loans	256	86.0	85.8	83.9	81.0
Purchasing	16	79.8	82.9	84.2	81.0
Purchasing	64	80.6	85.5	85.6	80.8
Purchasing	256	84.5	84.6	86.0	81.7

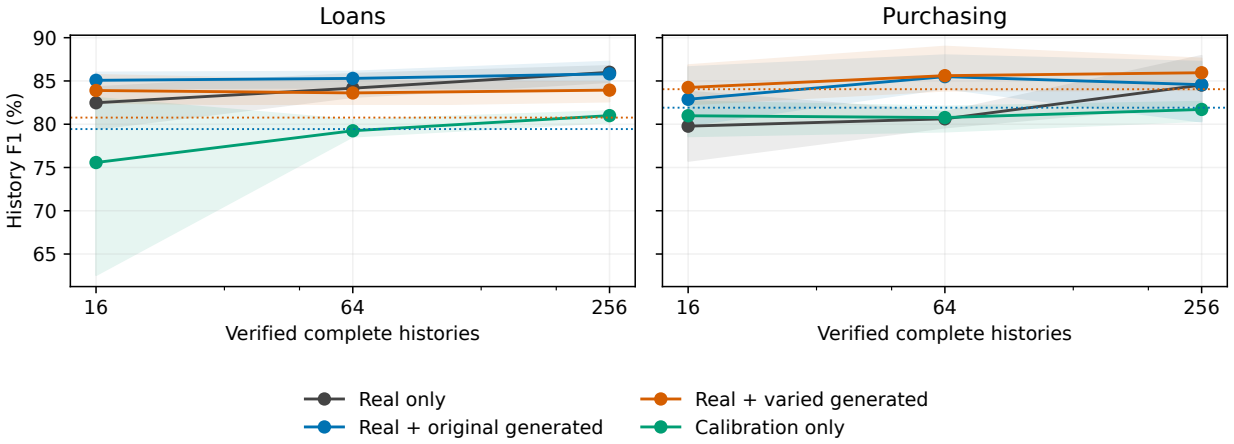


Figure 2: History recovery against complete verified-history budget. Lines show three-seed means; bands show observed minima and maxima, not confidence intervals. Dotted lines show fixed generated scorers without new target verification.

5.3 Operational recovery and introduced harm

At budget 64, the purchasing varied mixture raises exact-history recovery from 36.5% to 48.7% relative to real-only fitting. It restores a mean of 3,880 previously incorrect event memberships to full correctness, versus 2,992, while reducing newly damaged events from 411 to 237. The unit is an event with its complete target-participation set, including contamination through merged aliases; these counts do not imply that every incorrect association on other events was unchanged.

Operational queries show a related but distinct benefit. Purchasing cancellation-evidence F1 rises from 89.3% to 91.1%; correct first and last timestamps rise from 73.2/73.2% to 81.2/79.2%. For loans, the original mixture improves cancellation-evidence F1 from 79.4% to 87.3% at budget 64, but correct first timestamps fall from 82.5% to 78.6%. A higher overall history score consequently does not imply improvement for every investigation question.

Identity recovery remains limited. All loan models fitted with budgets 16 or 64 recover no true alias pairs under their calibrated operating points. On purchasing, the varied mixture at budget 64 recovers true pairs in only one trial. At budget 256 it reaches mean identity-pair precision .912 and recall .378. The supported edit vocabulary is broader than the reliable recovery achieved at small verification budgets.

Tightening the empirical calibration precision target reduces mean new damage in all four tested domain/regime combinations (Table 4). It also reduces complete event recovery in each. For real-

Table 4: Threshold sensitivity at budget 64. Columns compare .95/.99 empirical calibration precision targets; neither is a guarantee on new histories. F1 is percent; damage and repaired-event counts are means over three seeds.

Application	Method	F1 .95	.99	Damage .95	.99	Repaired .95	.99
Loans	Real only	84.2	85.7	671.7	227.3	4891.0	4319.7
Loans	Real + varied generated	83.6	83.1	756.3	278.0	3994.7	2566.3
Purchasing	Real only	80.6	79.6	411.3	84.7	2992.0	1217.7
Purchasing	Real + varied generated	85.6	83.2	237.3	154.3	3880.3	2613.0

Table 5: Mean history F1 (percent) and newly damaged event counts under clean inputs and independent pipeline faults. Adapted models use 64 verified bundles; generated-only models use zero new target bundles.

Application	Method	Clean F1	Damage	Pipeline F1	Damage
Loans	Real only	99.6	65.7	77.9	680.7
Loans	Real + original generated	99.6	111.0	78.4	722.0
Loans	Real + varied generated	99.9	53.0	77.8	740.3
Loans	Generated original	100.0	12.0	77.7	194.0
Loans	Generated varied	100.0	28.0	77.4	448.3
Purchasing	Real only	100.0	25.7	77.9	337.7
Purchasing	Real + original generated	100.0	2.3	80.3	660.7
Purchasing	Real + varied generated	99.9	20.7	80.3	450.7
Purchasing	Generated original	100.0	2.0	79.0	193.3
Purchasing	Generated varied	99.9	24.0	79.5	428.0

only loans, stricter calibration improves history F1 to 85.7%, above the 85.3% original-mixture result at the default operating point. For purchasing, tightening the varied mixture lowers history F1 from 85.6% to 83.2%. Thus the default budget advantage should not be read as superiority at every harm tolerance. The atomic-root control remains a development alternative because its recovery/damage tradeoff was inconsistent relative to simple confidence tightening.

5.4 Clean records and a different fault process

Table 5 retains both clean-input and independent pipeline-fault outcomes. Every tested method introduces some damage across its clean-record trials, although some individual runs have zero damage; displayed history F1 values rounded to 100.0% do not mean zero errors. Under pipeline faults, mean damaged history F1 is 77.2% on loans and 77.7% on purchasing. The best reported mixture means rise only to 78.4% and 80.3%, respectively. The purchasing original mixture introduces more new damage than real-only fitting under this stressor despite higher history F1. These results support cautious use as auditable repair proposals; a calibrated local edit score does not certify the correctness of a complete repaired history.

5.5 Compatible published-method comparison

The evaluation masks each event's case identifier independently with probability .30 and restricts predictions to the closed catalog of known cases. Candidate recall averages 98.1% (range 98.1–98.2%) for loans and 86.0% (range 85.9–86.0%) for purchasing. Generated supervision has no universal advantage: both generated methods exceed nearest-event accuracy on loans, while gener-

Table 6: Native missing-identifier accuracy across three seeds. Entries report mean (min–max) percent.

Method	Loan (BPI 2017)	Purchasing (BPI 2019)
Generated, original	89.8 (89.5–90.1)	8.3 (7.8–8.9)
Generated, varied	88.9 (88.4–89.2)	6.1 (5.5–6.7)
Known-label HGB	94.5 (94.4–94.6)	65.1 (64.7–65.7)
Nearest event	83.7 (83.3–84.0)	59.0 (58.8–59.3)
HERE-Transformer (no expert rules)	67.9 (66.5–69.0)	42.0 (41.5–42.3)

ated accuracy falls to 8.3% and 6.1% on purchasing, well below nearest event and the known-label HGB control. The known-label HGB control has the highest accuracy in both domains. HERE-Transformer is evaluated without expert rules. These results limit the role of generated-only transfer. High loan accuracy does not carry over to purchasing after the input is reduced to activity, resource and time. Candidate coverage exceeds 85% in both applications, so retrieval failure alone cannot explain the purchasing losses. The strongest control in this experiment learns from trusted retained case IDs. Its advantage does not establish superiority on incorrect existing links or fragmented identities, which this experiment does not introduce.

5.6 What a repair changes for an investigator

Figure 3 illustrates both useful recovery and harm in the earlier loan study. In the successful example, the retrieved application history loses an unrelated call and gains its missing offer-creation record. Its recorded cancellations remain visible. The second example shows why aggregate improvement is insufficient: repair removes the only correctly retrieved event of a short offer history while leaving an unrelated event. An investigator would then receive a less faithful account of that offer. These are post hoc illustrations from one original trial, selected by documented rules, not additional independent validation or estimates of how often each outcome occurs.

6 Scope, limitations and reproducibility

The earlier paired study supports observation variation as a useful generated-data condition, but synthetic augmentation does not always beat verified operational supervision. The purchasing budget comparison supports a reduction in incremental verified bundles within one previously studied process. It is not a general sample-complexity result, a measured reduction in human work, or evidence of transfer to an unseen institution. The fixed mixture weighting was not tuned separately for each source; better adaptation might alter the observed boundaries, but was not tested here.

All operational error labels in the main evaluation come from controlled perturbations of released representations. Public natural-error candidates were audited without substituting their tasks: S2AND identity disagreements largely require splitting overmerged profiles, while the public object-centric imperfection demonstration compares object–object relationships across logs. Neither provides independent natural business-error ground truth compatible with the implemented operations. Field validation with verified operational corrections remains open.

Sparse or empty histories may not contain enough information to identify a missing participant. False merges can contaminate many retrieved histories. Confidence thresholds fitted on a few bundles are uncertain, and precision measured on those bundles is not a deployment guarantee.

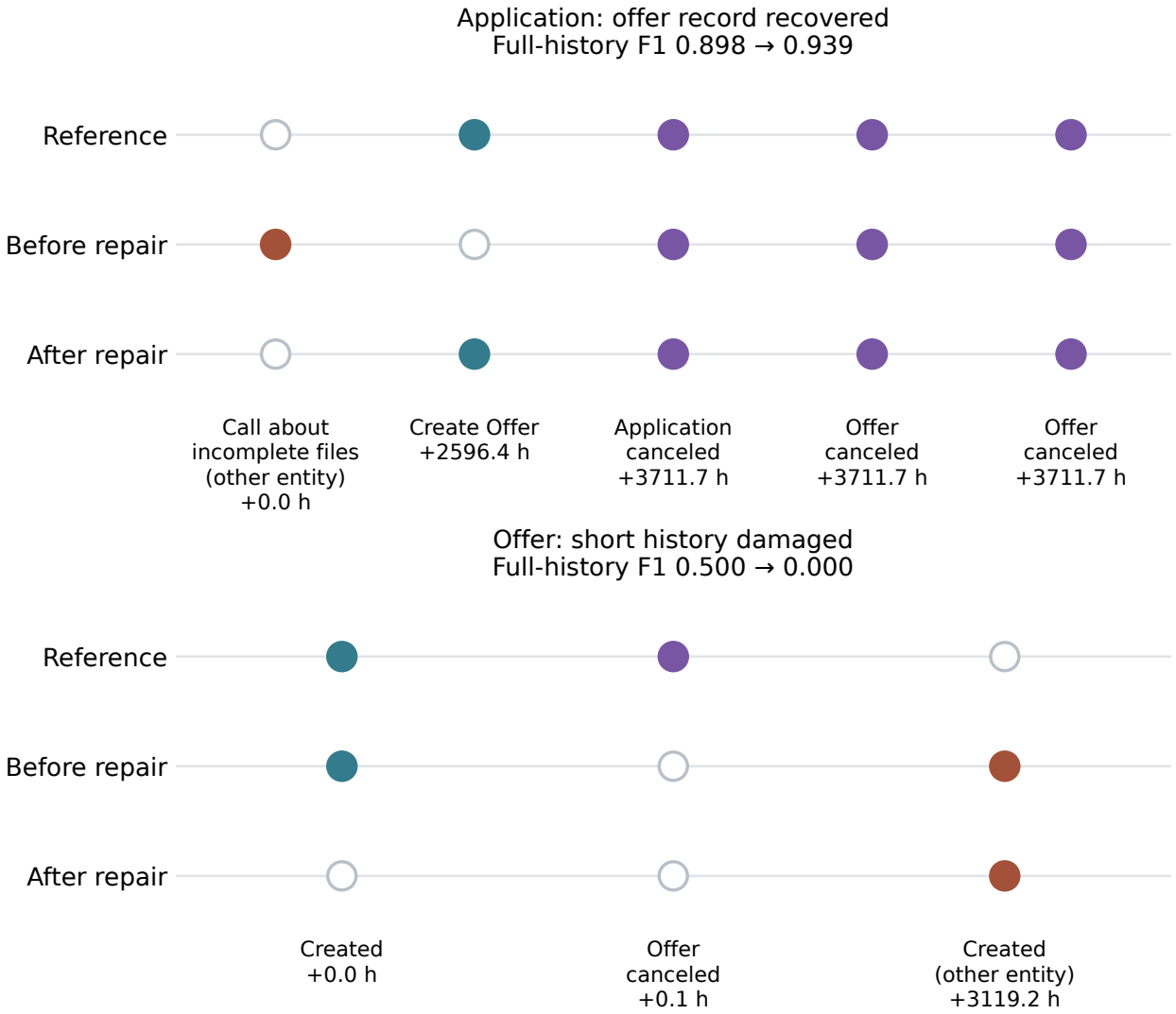


Figure 3: Application consequences in the earlier controlled-fault loan experiment. Filled markers denote retrieved records; empty markers denote absent records. Purple marks cancellation evidence, blue other correct records, and brown unrelated records. The upper panel shows changed associations and cancellation records from a 26-record reference history; the lower shows the full three-record union for a two-record reference history. Displayed F1 scores use complete histories. Columns are chronological but evenly spaced; elapsed hours are labelled.

The fault families, uncorrupted context, available catalog and offline window all limit transfer claims. Sampling complete roots also changes concurrency relative to a complete enterprise log.

The original study included a BPI 2012 check [15], architecture controls and generated compositional tests. Their results and failures remain in the separate supporting artifact. Four originally declared external simulation sources could not be retrieved; their planned evaluations remain explicitly unexecuted. No substitute was chosen based on outcomes. The revision preserves that earlier study and does not present it as untouched confirmation for newly selected methods.

Recipes, split rules, label-access tests, source/model hashes, predictions, edit audits and per-trial results support reproduction. All 156 combined-repair outputs passed saved-edit replay, event-

content preservation and independently implemented primary-metric checks. The native accuracy audit reconstructs truth directly from the pinned source memberships. Both domains' four original revision partitions were rebuilt byte-identically from their public sources. Refitting six selected budget-64 models reproduced calibration decisions and thresholds, with score differences below 1.4×10^{-13}; model-file byte identity did not reproduce. These checks distinguish stored-model replay from fresh retraining. No human annotation-time study or independent hardware replication is claimed.

7 Conclusion

Executable temporal workflows can provide useful repair supervision when verified operational histories are scarce. The strongest fresh result is application-specific: on purchasing, the varied-generated mixture with 64 verified bundles exceeds real-only fitting with 256 bundles in every paired trial. Observation variation alone is insufficient to select a reliable repair system. Loans favor a different mixture, a strong local control leads the separate missing-ID comparison, and even nearly perfect aggregate scores on clean data conceal newly damaged events. The practical outcome is a reproducible way to evaluate generated repair supervision against verification cost and introduced harm. Independent natural-error validation and dependable identity recovery remain necessary before unattended operational use.

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